Exact, hardware-aware local rewriting of quantum circuits, with certified ground truth

Arnav Kapoor[1*]

[1*]Indian Institute of Science Education and Research Bhopal, Bhopal, Madhya Pradesh, India.

Corresponding author(s). E-mail(s): kapoorarnav43@gmail.com;

Abstract

Local rewriting shortens a quantum circuit by replacing short sub-circuits with cheaper equivalents from a precomputed database. Building on the database method of Rosenhahn, Osborne and Hirche, we turn local rewriting into an exact, hardware-aware optimiser and give it a mathematical foundation. Every notion of equivalent sub-circuit we need is the orbit relation of a group action, which fixes the database keys and yields soundness and error bounds. The optimiser is gate-set agnostic, exact including global phase, uses calibrated costs and connectivity-aware convex windows, relocates circuits to the cheapest region of the chip, and emits replay certificates that an independent checker verifies a few qubits at a time. For discrete gate sets we prove when local optimality is global, enumerate all 92,897,280 three-qubit Clifford elements and certify optimal lengths of two-qubit Clifford+T circuits in exact arithmetic. Against this ground truth the optimiser reaches the proven optimum on 17/20 and 18/20 three-qubit benchmark circuits, where the earlier database method is **4.5**× and **3.0**× above it, and on 41/41 planted Clifford+T circuits. On 58 algorithm circuits compiled by Qiskit for IBM Heron it removes 13% of the gates without adding a two-qubit gate, after BQSKit it removes a further 16%, on circuits of up to 95 qubits it removes 19%, and run after VOQC it gives the best of six tools at the logical level. Every reported output is verified.

Keywords: quantum circuit optimization, peephole optimization, Clifford group, Clifford+T, Cayley graph, exact synthesis, qubit mapping, noise-aware compilation

1 Introduction

Compiling a circuit into a device's native gates leaves redundancy, and on noisy hardware every gate costs fidelity. Rosenhahn, Osborne and Hirche [1] showed that a strikingly simple, gate-set-agnostic idea already shortens circuits beyond Qiskit [2]. A window of a few gates is replaced by a shorter equivalent word looked up in an enumerated database of small circuits. The idea belongs to the family of database-driven and peephole optimisers [3–10], but needs no hand-written rules.

This paper takes the approach from a proof of concept to an optimiser for real devices, and supplies the theory and the ground truth needed to judge it. Its guiding principle is a strict separation between a *general optimiser*, which uses no structure of the gate set, and *exact ground truth*, which does and is used only for evaluation.

Contributions.

1. *Foundation.* Equivalence notions are group actions, database keys are canonical orbit representatives, and soundness, composition and an additive error bound follow (Section 3).
2. *Optimiser.* Exact rewriting including global phase, calibrated costs, continuous angles, connectivity-aware convex windows, layout-aware relocation and replay certificates (Sections 4 and 5).
3. *Theory and certified ground truth.* An exactness threshold and cascade, a parity theorem, mixing of random benchmarks and counting lower bounds (Section 6), complete distance tables of the 3-qubit Clifford group and exact two-qubit Clifford+T distances with planted benchmarks (Section 7).
4. *Evaluation* on the benchmarks of [1] against proven optima, on 58 algorithm circuits for IBM Heron and trapped ions, on circuits of up to 95 qubits, at the logical level and on an IBM device, against Qiskit, TKET [11], BQSKit [12], VOQC [13] and QUESO [9] (Section 8).

The implementation, `qcr`, and every script and raw result are openly available (see Declarations).

2 Related work

Local and database-driven optimisation.

Peephole optimisation with databases of optimal small circuits goes back to reversible logic [3, 4]. For quantum circuits, rule-based optimisers with continuous parameters [5, 13], template matching [6, 7] and superoptimisers that turn enumerated equivalent pairs into rewrite rules [8–10] are standard. The method of [1] stores optimal words directly and needs no rule generation. We keep this simplicity and change what is stored (canonical orbit keys), where it is applied (convex windows on coupled qubits) and what is optimised (calibrated cost). Resynthesis compilers such as BQSKit [12] re-synthesise blocks numerically [14, 15], exact only up to a tolerance.

Optimal synthesis and ground truth.

Clifford circuits are described by stabiliser tableaux [16], have a complete set of relations [17] and are known optimally up to six qubits [18]. Clifford+T unitaries are those with entries in $\mathbb{Z}[1/\sqrt{2}, i]$ [19], and normal forms and meet in the middle give optimal circuits for small instances [20–24]. We use these ingredients to certify optimal lengths over a native hardware alphabet, in the spirit of known-optimal benchmarks for qubit mapping [25].

Layout and noise-aware compilation.

Routing [26] and noise-adaptive mapping [27, 28] choose where a circuit runs, and Qiskit's VF2 layout passes [2, 29] score placements by calibrated errors. Our relocation scores placements after rewriting and includes readout errors.

3 Framework

3.1 Devices, circuits and costs

Definition 1 (Device and cost) A device is a tuple $\mathcal{D} = (Q, E, \Gamma_1, \Gamma_2, e)$ of physical qubits Q, a coupling graph $E \subseteq \binom{Q}{2}$, native single-qubit gates Γ_1 (possibly with continuous angles), native two-qubit gates Γ_2 allowed on pairs in E, and error rates $e(\gamma) \in [0, 1)$ of every gate on its physical qubits and $e_{\mathrm{m}}(q)$ of every measurement. A circuit is a word $u = \gamma_1 \cdots \gamma_L$ of native gates, and $\mathrm{ev}(u) = \gamma_L \cdots \gamma_1$. An additive cost is $c(u) = \sum_i c(\gamma_i)$ with

$$c_{\mathrm{count}} \equiv 1, \qquad c_{w2}(\gamma) = \begin{cases} 10 & \gamma \in \Gamma_2 \\ 1 & \gamma \in \Gamma_1 \end{cases}, \qquad c_{\mathrm{cal}}(\gamma) = -\log\bigl(1 - e(\gamma)\bigr). \tag{1}$$

The calibrated cost is the negative logarithm of the estimated success probability (ESP) [30],

$$\mathrm{ESP}(u) = \prod_{i=1}^{L} \bigl(1 - e(\gamma_i)\bigr) \prod_{q \in A(u)} \bigl(1 - e_{\mathrm{m}}(q)\bigr) = \exp\Bigl(-c_{\mathrm{cal}}(u) - \sum_{q \in A(u)} c_{\mathrm{m}}(q)\Bigr), \tag{2}$$

with $A(u)$ the measured qubits and $c_{\mathrm{m}}(q) = -\log(1 - e_{\mathrm{m}}(q))$. We use two device families. *IBM Heron* has $\mathrm{RZ}(\theta)$ (virtual, $e = 0$), SX, X and CZ on a heavy-hex graph with calibrated errors from Qiskit's snapshots (FakeTorino, 133 qubits, couplers with error 1.0 removed). The *trapped ion* model has all-to-all $\mathrm{RX}, \mathrm{RY}, \mathrm{RZ}(\theta)$ and $\mathrm{RXX}(\theta)$ [31] with $e = 3 \cdot 10^{-5}$ (single-qubit), 10^{-3} (two-qubit) and virtual RZ.

Definition 2 (Convex window) The DAG of u has vertices $1, \ldots, L$ and an edge $i \to j$ whenever $i < j$ and γ_i, γ_j act on a common qubit. A set S of vertices is *convex* if every directed path between two vertices of S stays in S. A window is a convex S whose wires $W(S)$ satisfy $|W(S)| \leq 3$ and, on a device with coupling graph E, induce a connected subgraph of E.

Table 1 Equivalence notions as group actions. Notions compose, e.g. output-aware equivalence modulo chip symmetries uses row-normalised matrices minimised over the automorphism group.

notion	$U \sim V$ iff	group H	key $\kappa_H(U)$
exact	$V = U$	trivial	U
up to phase [1]	$V = e^{i\varphi}U$	$U(1)$	phase-normalised U
input-aware	$V\|0\rangle = e^{i\varphi}U\|0\rangle$	stabiliser of $\|0\rangle$	phase-normalised $U\|0\rangle$
output-aware	$V = DU$, D diagonal	diagonal unitaries	rows phase-normalised
symmetry	$V = PUP^\dagger$	graph automorphisms	minimum over the orbit
approximate	$\|U - V\| \le \varepsilon$	(a tolerance)	exact key, checked distance

Two gates are *independent* if they commute exactly, and words equal up to swapping adjacent independent gates are trace-equivalent [32, 33]. The factors of trace-equivalent words are exactly the convex sets, so a convex window can be moved to be contiguous without changing $\mathrm{ev}(u)$. Random commuting shuffles as in [1] enumerate convex windows stochastically, and we enumerate them directly.

3.2 Equivalence as a group action

Definition 3 (Equivalence notion) An equivalence notion on k qubits is a group H acting on $U(2^k)$, with $U \sim_H V$ if and only if $V \in H \cdot U$. A *canonical key* is a map κ_H with $\kappa_H(U) = \kappa_H(V) \iff U \sim_H V$.

Table 1 lists the notions we use. A database stores, for each key, a cheapest word, so one lookup of $\kappa_H(\mathrm{ev}(S))$ finds every member of the orbit. Exact equivalence including global phase is the only notion that stays sound when the circuit is later controlled.

Proposition 1 (State- and measurement-aware rewriting) *Let the program be "prepare $|0^n\rangle$, apply $\mathrm{ev}(u)$, measure in the computational basis", and let S be a window of u on wires W. (a) If no gate precedes S on W and $\mathrm{ev}(v)|0_W\rangle = e^{i\varphi}\mathrm{ev}(S)|0_W\rangle$, replacing S by v preserves the output distribution. (b) If no gate follows S on W and $\mathrm{ev}(v) = D\,\mathrm{ev}(S)$ with D diagonal, replacing S by v preserves the output distribution. Any sequence of such rewrites, each applied to the current circuit, preserves the output distribution.*

Proposition 2 (Approximate rewriting) *If rewrites $t = 1, \ldots, m$ replace windows S_t by v_t with $\|\mathrm{ev}(v_t) - \mathrm{ev}(S_t)\| \le \varepsilon_t$, the final circuit satisfies $\|\mathrm{ev}(u') - \mathrm{ev}(u)\| \le \sum_t \varepsilon_t$ (including global phase in exact mode).*

All proofs are in Appendix A. The optimiser reports $\sum_t \varepsilon_t$ with every circuit. Its tests use the norm distance, since a fidelity test $|\mathrm{tr}(U^\dagger V)|/2^k \ge 1 - \mathrm{tol}$ admits errors of order $\sqrt{\mathrm{tol}}$.

3.3 Rewriting systems

Definition 4 (Rewriting system) For a window bound w, a database radius r, an equivalence notion H and a cost c,

$$\mathcal{R}(w,r) = \big\{u \to v \;\big|\; |u| \le w,\ |v| \le r,\ \mathrm{ev}(u) \sim_H \mathrm{ev}(v),\ c(v) < c(u)\big\}. \tag{3}$$

A word is (w,r)-*reduced* if no trace-equivalent word has a factor that is the left side of a rule.

For a finite alphabet Γ (modulo phase) let $G = \langle\Gamma\rangle$, $|g|$ the length of a shortest word for g, $B(r)$ and $S(r)$ the ball and sphere of radius r in the Cayley graph $\mathrm{Cay}(G,\Gamma)$, and $D = \max_g |g|$ the diameter. The compute graph of depth d of [1] is $B(d)$, and its optimal factorisations are geodesics.

4 The optimiser

Selection rule.

Let $\mathcal{W}(u)$ be the windows of u (Definition 2) grown from each position by collecting later gates that commute exactly (tolerance 10^{-10}) past the gates they jump over. For a window S let $v(S)$ be the cheapest replacement found by the lookups below. A sweep applies

$$S^\star = \arg\max_{S \in \mathcal{W}(u)} \Delta(S), \qquad \Delta(S) = c(S) - c\big(v(S)\big), \tag{4}$$

over every prefix of every window, while $\Delta(S^\star) > 0$. Sweeps alternate with random exact-commutation shuffles until p consecutive shuffles bring no gain. Since every step lowers c, the output never costs more than the input.

Databases.

For each device we fix a grid of native gates on a three-qubit local topology (all pairs for ions, the line 0–1–2 for heavy-hex). The Heron grid has $\mathrm{RZ} \in \{\pm\frac{\pi}{4}, \pm\frac{\pi}{2}, \pi\}$, SX, X and CZ (23 local gates), the ion grid $\mathrm{RX}, \mathrm{RY}, \mathrm{RZ}$ on the same angles and $\mathrm{RXX}(\pm\frac{\pi}{4}, \pm\frac{\pi}{2})$ (57 local gates). Words are enumerated breadth first, and each new class is stored under a 64-bit hash of $\min_{\pi \in \mathrm{Aut}} \kappa_H(\pi\,\mathrm{ev}(v)\,\pi^\dagger)$. Matrices are kept only below the deepest level, and every hit is re-checked numerically. Table C5 lists the databases and Fig. 1(b) their growth. They are not told that any group is finite or Clifford.

Lookup beyond the radius.

If $\kappa(\mathrm{ev}(S))$ is not stored, meet in the middle searches $A \in B(2)$ with $\kappa(\mathrm{ev}(S)A^{-1})$ stored, so that $|S| \le |\mathrm{ev}(S)A^{-1}| + |A|$ is realised by a stored word followed by A.

Continuous angles.

Windows on at most two coupled wires are also resynthesised analytically, on Heron by the KAK decomposition with the minimal number of CZ [34], on ions by a Weyl-chamber decomposition [35] with one $\mathrm{RXX}(\theta)$ per non-zero interaction coefficient, and single-qubit runs by Euler decompositions.

Global phase.

Inside windows the optimiser matches up to phase and records the phase φ_t of each replacement, snapped to the set Φ the device realises exactly ($\Phi = \frac{\pi}{4}\mathbb{Z}$ on Heron, $\frac{\pi}{2}\mathbb{Z}$ on ions). At the end the total $\varphi = \sum_t \varphi_t \bmod 2\pi$ is repaired with the cheapest gadget, or on ions by $\mathrm{RXX}(\theta \pm \pi)\,\mathrm{RX}(\pi)^{\otimes 2} = \pm i\,\mathrm{RXX}(\theta)$. The output equals the input including global phase.

Replay certificates.

Statevector checks end at about 25 qubits, and the decision-diagram checker of [36] returned no result on routed 23-qubit adders. The optimiser therefore logs every accepted step.

Definition 5 (Replay certificate) A certificate for $u \mapsto u'$ is a sequence of steps $u = u_0 \to u_1 \to \cdots \to u_m = u'$, each either a swap of adjacent gates γ, γ' or a replacement of a contiguous factor s by v with a scalar λ_t. The checker accepts it if

$$\|[\gamma, \gamma']\| \le \delta_t \text{ (swaps)}, \qquad \|\mathrm{ev}(v) - \lambda_t \mathrm{ev}(s)\| \le \delta_t \text{ (replacements)}, \qquad \Big|\prod_t \lambda_t - 1\Big| \le \delta_0, \quad (5)$$

with all matrices formed on the step's own wires only.

Proposition 3 (Certificates) *If a certificate is accepted,* $\|\mathrm{ev}(u') - \mathrm{ev}(u)\| \le \delta_0 + \sum_t \delta_t + O(\delta^2)$ *with* $\delta = \max_t \delta_t$.

The checker shares no databases or rules with the optimiser and builds only few-qubit matrices, so its cost grows with the number of steps, not with the width.

5 Layout-aware rewriting

Rewriting cannot change where a circuit runs, yet the ESP of one circuit can differ by an order of magnitude between placements.

Definition 6 (Relocation) Let u be a routed circuit on physical qubits A with interaction graph $I(u)$. A relocation is an injective $\phi : A \to Q$ with $\{\phi(a), \phi(b)\} \in E$ for every edge $\{a, b\}$ of $I(u)$, and u^{ϕ} relabels every gate by ϕ.

Relabelling preserves the output distribution up to relabelling the measured qubits, so relocation is exact. We choose

$$\phi^{\star} = \arg\min_{\phi} \Big(c_{\mathrm{cal}}(u^{\phi}) + \sum_{q \in A} c_{\mathrm{m}}(\phi(q)) \Big) = \arg\max_{\phi} \ \mathrm{ESP}(u^{\phi}), \quad (6)$$

enumerating subgraph monomorphisms with VF2 [29] anchored at every physical qubit. The identity is a candidate, so relocation never lowers ESP. *Joint layout selection* optimises and relocates several candidate layouts (Qiskit level-3 transpilations

with several seeds, with and without the calibrated target) and keeps the cheapest, since the best layout before optimisation need not be the best after it.

6 Theory for discrete gate sets

This section concerns the alphabets of [1], $\Gamma_{\text{ion}} = \{\mathrm{RX}, \mathrm{RY}, \mathrm{RZ}(\pm\frac{\pi}{2})\} \cup \{\mathrm{RXX}(\pm\frac{\pi}{2})\}$ and $\Gamma_{\text{ibm}} = \{\mathrm{RX}, \mathrm{RZ}(\pm\frac{\pi}{2})\} \cup \{\mathrm{CZ}\}$, modulo global phase.

Lemma 4 (Clifford structure) *$\langle\Gamma\rangle = \mathcal{C}_n/U(1)$, of order $2^{n^2+2n}\prod_{j=1}^{n}(4^j-1)$, and the compute graph of depth d is the ball $B(d)$ of $\mathrm{Cay}(\mathcal{C}_n/U(1), \Gamma)$.*

6.1 When local optimality is global

Theorem 5 (Exactness threshold) *Let G be finite with diameter D. If $w \geq D+1$ and $r \geq D$, every (w, r)-reduced word is a geodesic.*

Proposition 6 (Cascade) *Let $\ell(w)$ be the fixed-point length of an exhaustive $\mathcal{R}(w, \infty)$ optimiser. If the current length is at most w at any time, the next sweep returns a geodesic. Hence the relevant window is*

$$w^\star = \min\{w \mid \ell(w) \leq w\}. \tag{7}$$

For 3-qubit circuits we measure $w^\star \approx 12$ (ion) and 16 (IBM-style), well below $D + 1 = 18$. For a universal gate set $D = \infty$ and no finite window is exact, so ground truth must come from bounded search (Section 7.2).

Theorem 7 (Parity) *(a) For $n \in \{1, 2\}$ and both alphabets, $\mathrm{Cay}(\mathcal{C}_n/U(1), \Gamma)$ is bipartite, so every equivalent rewrite changes the length by an even number. (b) For $n \geq 3$ and any generating set of $\mathcal{C}_n/U(1)$ that contains, for each qubit, a gate squaring to X_i and one squaring to Z_i, the Cayley graph is not bipartite.*

For $n = 3$ exhaustive search gives a shortest odd relation of length $\rho = 11$ (ion) and 9 (IBM-style), so a rewrite that changes parity needs a window of at least $\lceil \rho/2 \rceil = 6$ gates.

6.2 What random-circuit benchmarks measure

Theorem 8 (Benchmarks mix) *Let $\Gamma = \Gamma^{-1}$, let μ_L be the law of $\mathrm{ev}(u)$ for u uniform in Γ^L, and let $\beta < 1$ be the largest modulus of a non-trivial eigenvalue of the random-walk operator (excluding -1 if the graph is bipartite). Then*

$$\|\mu_L - \pi_L\|_{TV} \leq \tfrac{1}{2}\sqrt{|G|}\,\beta^L, \qquad \mathbb{E}\,|\mathrm{ev}(u)| \xrightarrow{L\to\infty} \frac{1}{|G|}\sum_d d\,|S(d)|, \tag{8}$$

where π_L is uniform on G (or on the parity class of L).

Table 2 Exact Cayley-graph data of the Clifford groups generated by the alphabets of [1].

alphabet	n	$\lvert\Gamma\rvert$	$\lvert G\rvert$	diameter D	mean $\lvert g\rvert$ (uniform)
ion	2	14	11,520	10	5.89
IBM-style	2	9	11,520	11	6.92
ion	3	24	92,897,280	17	10.72
IBM-style	3	15	92,897,280	19	12.34

For $n = 2$ (ion) $\beta = 0.937$. A random benchmark of length 300 therefore has a constant optimal answer, 10.72 gates for 3-qubit ion circuits and 12.34 for IBM-style ones, for all $L \gtrsim 100$.

6.3 Lower bounds for every method

Proposition 9 (Counting bound) *For g uniform in G,*

$$\Pr\big[|g| \le d\big] \le \frac{|B(d)|}{|G|}, \qquad |S(a+b)| \le |S(a)|\,|S(b)|, \qquad \mathbb{E}|g| = \sum_{d\ge 0} \Pr\big[|g| > d\big], \tag{9}$$

so exact sphere sizes up to depth k give lower bounds on $\mathbb{E}|g|$ that hold for every optimiser.

At $n = 4$ ($|G| = 1.21 \times 10^{13}$), exact spheres to depth 6 give $\mathbb{E}|g| \ge 11.1$ (ion) and ≥ 12.9 (IBM-style).

7 Certified ground truth

7.1 The three-qubit Clifford group

A Clifford modulo phase is determined by its signed tableau [16]. For $n \le 3$ the tableau fits into one 64-bit word, gates act through lookup tables, and a breadth-first level is a vectorised gather. Table 2 and Fig. 1(a) summarise the complete enumeration (25 minutes and about 6 GB per alphabet). The numbers of 3-qubit elements needing exactly $0, 1, \ldots, 6$ entangling gates are 13 824, 373 248, 5 971 968, 38 486 016, 42 052 608, 5 971 968 and 27 648 (mean 3.51), the same for both alphabets since $\mathrm{RXX}(\frac{\pi}{2})$ and CZ are locally equivalent.

7.2 Beyond Clifford: exact two-qubit Clifford+T distances

With angles in $\{\pm\frac{\pi}{4}, \pm\frac{\pi}{2}, \pi\}$ the ion alphabet ($\mathrm{RX}, \mathrm{RY}, \mathrm{RZ}$ on each qubit and $\mathrm{RXX}(\pm\frac{\pi}{4}, \pm\frac{\pi}{2})$, 34 gates on two qubits) is universal, and the group is infinite. Up to phase every generator has entries in $\mathbb{D}[\omega] = \mathbb{Z}[1/\sqrt{2}, \omega]$, $\omega = e^{i\pi/4}$, e.g.

$$\mathrm{RZ}(\tfrac{\pi}{4}) \propto \mathrm{diag}(1, \omega), \quad \mathrm{RX}(\theta) \propto H\mathrm{RZ}(\theta)H, \quad \mathrm{RXX}(\theta) \propto H^{\otimes 2}\mathrm{diag}(1, e^{i\theta}, e^{i\theta}, 1)H^{\otimes 2}, \tag{10}$$

and so does every circuit [19]. We store $U = M/\sqrt{2}^{\,k}$ exactly, with M over $\mathbb{Z}[\omega]$ in the $\mathbb{Z}$-basis $1, \omega, \omega^2, \omega^3$.

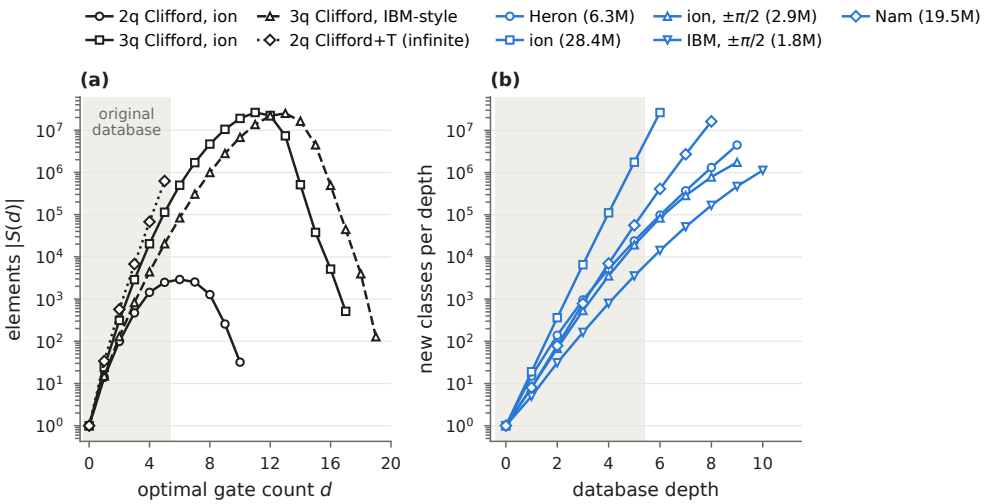


Fig. 1 (a) Sphere sizes $|S(d)|$ of the exact Cayley graphs used as ground truth. The shaded band is the database radius of [1]. The Clifford+T spheres keep growing, since that group is infinite. (b) New classes per depth in the general optimiser's databases (equality up to phase, one curve per device and alphabet, total size in parentheses).

Lemma 10 (Exact projective canonical form) *For U over $\mathbb{D}[\omega]$ let p be the first non-zero entry in row-major order and $P(U)_{ij} = U_{ij}\overline{U_p}$. Then $P(U)$ is over $\mathbb{D}[\omega]$, and $V = e^{i\varphi}U$ for some φ if and only if V has the same first non-zero position and $P(V) = P(U)$. Writing $P(U) = M'/\sqrt{2}^{k'}$ with the smallest $k' \geq 0$ for which M' is over $\mathbb{Z}[\omega]$ makes (M', k') unique.*

We hash the canonical forms into 64-bit keys and grow a ball of radius $R = 5$ (698,512 elements, levels 1, 34, 569, 6,744, 67,664, 623,500, 8 minutes). Equality of the numbers of distinct keys and distinct canonical forms certifies that no collision occurred. For $|U| \leq R + R'$ meet in the middle gives the exact distance

$$|U| = \min_{0 \leq j \leq R'} \Big(j + \min_{Y \in S(j)} \big|UY^{\dagger}\big| \Big), \tag{11}$$

which with $R' = 5 - 1$ certifies optimal lengths up to 9 and a lower bound beyond.

Planted benchmarks.

Long circuits are beyond any certifiable radius, so we plant the optimum. Starting from a core c with certified $|\mathrm{ev}(c)| = |c|$, we repeatedly pick a window W (up to four gates) and a word r (up to three gates), compute a geodesic s of $\mathrm{ev}(W)\mathrm{ev}(r)^{-1}$ exactly and replace W by rs. Since $\mathrm{ev}(s)\mathrm{ev}(r) = \mathrm{ev}(W)$ the unitary, and hence the certified optimum $|c|$, is unchanged, while overlapping windows make the planted structure non-local.

Table 3 Mean gate counts on the discrete benchmarks. *Original and Qiskit at 4 qubits use the first 30 circuits. Cliff. CW uses adaptive convex windows with the exact Clifford tables (Appendix B), which the general optimiser does not use. Optimum is the proven optimum or a certified lower bound, and #opt. counts circuits on which the general optimiser is optimal. All 250/250 outputs verified and native.

setting	Qiskit	orig.	Cliff. CW	general	optimum	#opt.
T3, ion, 3q	127.2	47.1	–	10.8	10.4	17/20
T4, IBM-style, 3q	104.0	36.8	–	12.5	12.2	18/20
journal, ion, 4q	164.9*	78.8*	45.6	46.1	≥11.1	–
journal, IBM-style, 4q	140.9*	76.0*	49.3	51.4	≥12.9	–
T5, IBM-style, 15q	332.1	335.9	324.6	327.7	–	–

8 Experiments

Metrics.

For a pipeline P and a set $\mathcal{C}$ of circuits, count ratios are ratios of sums and ESP ratios are geometric means,

$$\rho_{\#}(P) = \frac{\sum_{c\in\mathcal{C}} \#_P(c)}{\sum_{c\in\mathcal{C}} \#_{\mathrm{L3}}(c)}, \qquad \rho_{\mathrm{ESP}}(P) = \Big(\prod_{c\in\mathcal{C}} \frac{\mathrm{ESP}_P(c)}{\mathrm{ESP}_{\mathrm{L3}}(c)}\Big)^{1/|\mathcal{C}|}, \tag{12}$$

relative to Qiskit level 3 (L3) unless stated otherwise.

Verification.

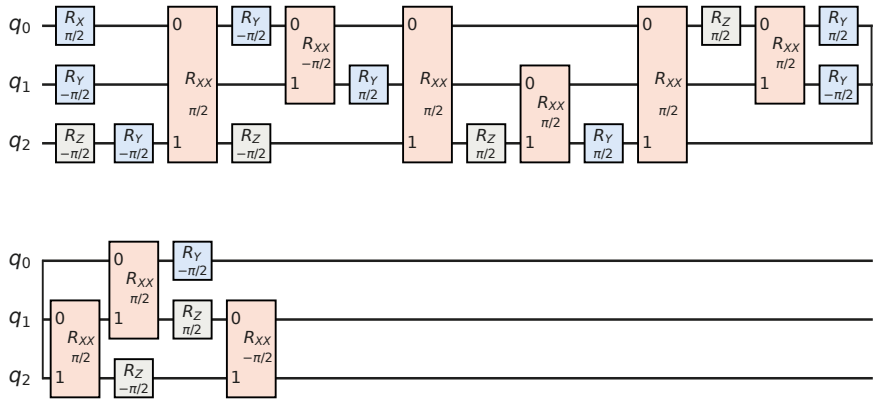
Every output is checked against its input. Exact mode is checked by $\|(U-V)\psi\|$ on random states (maximum $8.2\cdot10^{-11}$), state-aware mode by the TVD of the output distributions on $|0^n\rangle$ (maximum $6\cdot10^{-12}$), approximate mode against the tracked bound, and Clifford+T outputs by exact equality over $\mathbb{D}[\omega]$. Every output is also checked to be native and to use only coupled pairs. All 250 benchmark outputs, all 720 hardware-suite outputs and all Clifford+T and layout outputs pass. BQSKit outputs are checked the same way, up to global phase and its final qubit permutation. Large-suite outputs are checked by replay certificates, and up to 24 active qubits also by statevector. Logical-level outputs of every tool are checked up to global phase, by statevector up to 20 qubits and by [36] above.

8.1 Discrete benchmarks against ground truth

We run the general optimiser with the semantics of [1] (up to phase, gate count, $\pm\frac{\pi}{2}$ alphabets, all-to-all) on their settings and seeds, Tables III/IV (3 qubits, 300 gates, 20 circuits each), the journal 4-qubit statistics (100 circuits) and Table V (15 qubits, 500 gates, 10 circuits). "Original" is the database method (V2) of [1], run from the authors' released code ported to Python.

Without knowing that the group is finite or Clifford, the general optimiser is within 0.4 gates of the proven optimum on average and optimal on 35 of 40 three-qubit

(a) original method, 24 gates



(b) proven optimum from the exact Cayley table, 11 gates

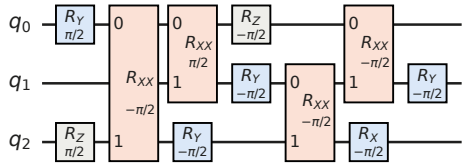


Fig. 2 One random 3-qubit word of 80 gates over Γ_{ion}, after the original method (a) and at its proven optimum (b). Both panels are drawn at the same scale.

circuits, where the original method is 4.5× and 3.0× above it (Table 3, Figs. 2 and C2). Fig. 3 explains the gap. At $w = 8$, raising r from 5 to ∞ lowers the mean excess from 46 to 18 gates, enlarging w at $r = 5$ barely helps, and for $w \geq 12$, $r \geq 8$ the excess is at most 2, as Theorem 5 predicts. At four qubits the general optimiser matches the Clifford-specific one, so large databases and convex windows, not knowledge of the group, make the difference. At 15 qubits all methods, Qiskit included, lie within 4% of one another.

8.2 Certified Clifford+T benchmarks

On the two-qubit Clifford+T alphabet we compare the general optimiser (ion database over exactly this alphabet), the original method (compute graph of depth 4 over the same alphabet) and Qiskit level 3 (continuous angles, shown for reference) with the certified optimum.

On random words with certified optimum the general optimiser is optimal on 99/99 circuits and the original method on 81/99 (Fig. 4). On 41 planted circuits (mean input 90.3, mean optimum 6.1) the general optimiser reaches the optimum on 41/41 (mean 6.1), the original method on 40/41 (mean 6.1), and Qiskit returns 7.7 gates in its continuous alphabet. All discrete outputs are exactly equal to their inputs over $\mathbb{D}[\omega]$ (example in Fig. C3).

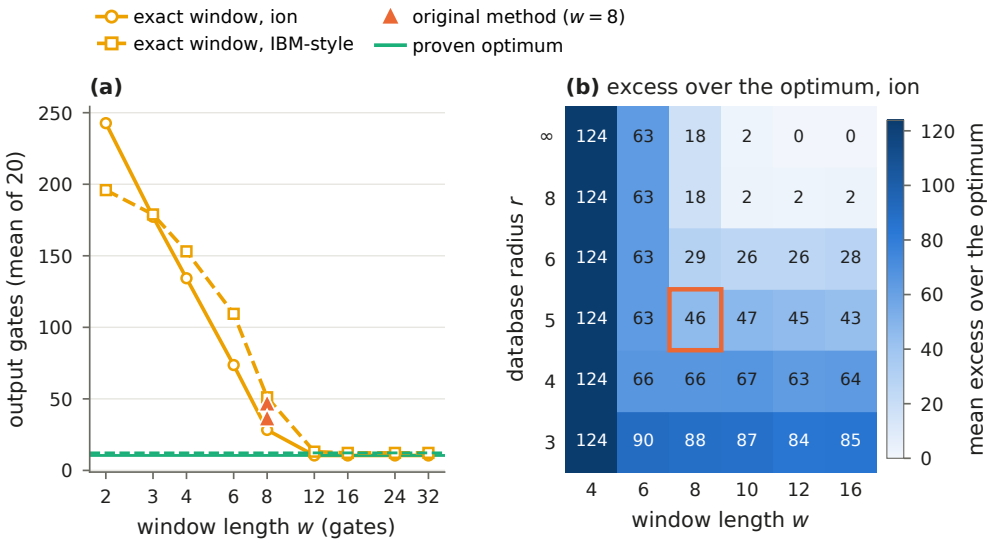


Fig. 3 Where the gap to the optimum comes from (3 qubits, exact Clifford tables). (a) Fixed-point length of the exact-window optimiser against the window length w, with the cascade near $w^\star$ of (7). (b) Mean excess over the optimum by window w and database radius r (ion), with the operating point $(8, 5)$ of [1] outlined.

8.3 Algorithms on hardware models

Suite.

60 circuits of 2–12 qubits and up to 1008 gates from Nam/VOQC arithmetic [5], QASMBench [37] (without feed-forward), QFT, QAOA, Heisenberg Trotter steps, EfficientSU2, a Steane encoder and random circuits are compiled by Qiskit at levels 1 and 3 with the same coupling map and seed. TKET `FullPeepholeOptimise` and BQSKit [12] (version 1.2.1, levels 2 and 3, blocks of up to three qubits resynthesised to 10^{-8}) start from the level-1 output.

Results.

Stacked on level 3 the optimiser never adds a two-qubit gate (9 circuits better and 49 equal on Heron, 1 and 57 on ions) and never lowers ESP (Table 4, Fig. C4, examples in Figs. C5 and C6). On Heron it removes 13% of all gates, mostly SX/X pulses and virtual RZ. On ions the total can rise (28 circuits) because c_{cal} trades pulses for virtual RZ. From level 1 it beats level 3 on Heron in all gates and, for seed 0, in ESP, of which $\times 1.11$ is due to Qiskit's level-1 layout. State-aware rewriting gives the largest reductions but empties 5 (Heron) and 7 (ion) reversible circuits with a fixed output. The median run time is 10.4 s (Heron) and 3.0 s (ion) per circuit.

Robustness.

Over 5 transpiler seeds (1,624 verified outputs) the Heron ratios on level 3 are $\times 0.975 \pm 0.004$ (two-qubit) and $\times 0.868 \pm 0.005$ (all). From level 1 the gate ratio is equally

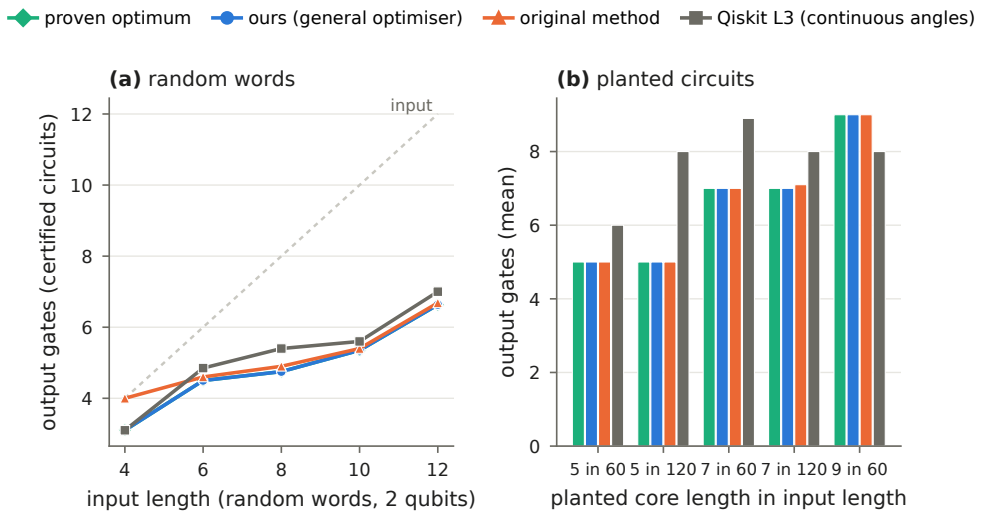


Fig. 4 Clifford+T, two qubits. (a) Random words, mean output length over circuits whose optimum is certified. (b) Planted circuits, a core of the stated length hidden in 60 or 120 gates.

Table 4 Ratios (12) relative to Qiskit level 3 (58 circuits per device). BQSKit outputs are approximate. The last row applies our optimiser to a separate BQSKit level-3 run. Nam pf1_10/pf2_10 are excluded because Qiskit level 3 drops their $\sim 10^{-7}$ rad rotations.

	IBM Heron			trapped ion		
pipeline	2q	all	ESP	2q	all	ESP
TKET (FullPeepholeOptimise on L1)	×1.02	×1.71	×1.16	×1.27	×1.79	×0.992
ours, exact, from L1	×1.00	×0.84	×1.21	×0.95	×0.76	×1.002
ours, exact, from L3	×0.98	×0.87	×1.03	×1.00	×1.00	×1.001
ours, approximate ($\varepsilon = 10^{-3}$), from L1	×1.01	×0.84	×1.21	×0.95	×0.75	×1.002
ours, state/measurement-aware, from L1	×0.77	×0.61	×1.28	×0.62	×0.50	×1.010
BQSKit level 2 on L1 (approximate)	×0.89	×1.26	×1.21	×0.96	×0.74	×1.002
BQSKit level 3 on L1 (approximate)	×0.85	×1.20	×1.22	×0.94	×0.71	×1.002
ours, exact, on BQSKit level 3	×0.85	×1.01	×1.25	×0.94	×0.70	×1.003

stable ($\times 0.844 \pm 0.006$), but the ESP gain ranges from ×0.97 to ×1.21 (mean ×1.10), following Qiskit's layout. Optimiser seeds change the total by at most 0.5%.

Resynthesis.

From the same level-1 input BQSKit level 3 uses fewer two-qubit gates than ours on 39 of 58 Heron circuits (more on 1) at about the same ESP, but 1.38 times as many gates as ours on level 3, with errors of median $1.9 \cdot 10^{-8}$ (maximum $1.4 \cdot 10^{-4}$), in 98 s per circuit on eight workers against 24 s on one. The two combine. On a BQSKit level-3 output (2459 two-qubit and 18601 total gates on Heron) our optimiser reduces

Table 5 Ablation on the hardware suite (58 circuits per device, seed 0), ratios (12) relative to Qiskit level 3. Each row changes one component of the full optimiser. All outputs verified.

	Heron, from L3			Heron, from L1		ion, from L1	
variant	saved	2q	ESP	all	2q	all	2q
full optimiser	13.0%	×0.98	×1.03	×0.84	×1.00	×0.76	×0.95
contiguous windows	11.7%	×0.99	×1.03	×0.88	×1.03	×0.82	×0.99
window $w = 8$	7.9%	×0.99	×1.02	×0.96	×1.17	×0.81	×0.97
window $w = 4$	1.3%	×1.00	×1.00	×1.09	×1.27	×1.06	×1.19
no shuffles	10.3%	×0.98	×1.03	×0.87	×1.02	×0.81	×0.98
no database	5.8%	×0.99	×1.02	×0.95	×1.04	×1.08	×1.01
database radius ≤ 3	6.2%	×0.99	×1.02	×0.94	×1.04	×0.89	×0.99
no meet in the middle	12.3%	×0.98	×1.03	×0.86	×1.02	×0.76	×0.95
no resynthesis	8.2%	×0.98	×1.02	×0.97	×1.15	×0.87	×1.29
gate-count cost	11.6%	×1.00	×1.01	×0.87	×1.11	×0.73	×1.08
2q-weighted cost	11.7%	×0.98	×1.02	×0.84	×1.01	×0.74	×0.95

the total to 15630 (×0.84), never adds a two-qubit gate and raises ESP on 54 circuits (×1.021), in 34 s per circuit. On ions BQSKit's output is already lean (×0.99).

8.4 What each component contributes

On Heron from level 3 the full optimiser saves 13.0% of all gates (Table 5). The window length matters most ($w = 8$ gives 7.9%, $w = 4$ gives 1.3%), then the database (none 5.8%, radius 3 6.2%), the same two parameters that set the gap in Fig. 3. Analytic resynthesis gives most of the rest (8.2% without it) and reduces the Mølmer–Sørensen count on ions (×1.29 of level 3's two-qubit gates without it, ×0.95 with it). Convex windows and shuffles add a little each and meet in the middle almost nothing (12.3%). Counting gates instead of c_{cal} saves as many gates but raises ESP only by ×1.010 instead of ×1.027.

8.5 Large circuits

45 Nam/VOQC circuits [5] of 14 to 100 qubits are compiled by Qiskit for the 133-qubit Heron model and an all-to-all ion model. Product-formula circuits (19 per device) are excluded as in Table 4, and so are circuits with an unchecked certificate (2 Heron, 1 ion), leaving 22 Heron circuits of 14–95 qubits and 25 ion circuits with up to 5,425 input gates. BQSKit ran on the 18 smallest circuits before it was stopped for time.

Stacked on level 3 the optimiser removes 19% of all gates and 2.1% of the two-qubit gates on Heron, removes two-qubit gates on 20 of 22 circuits, never adds one, and raises ESP by ×1.12 (median ×1.14 over all 22), with the same guarantees on ions (Table 6). From level 1 it ends with fewer gates than level 3 on every Heron circuit, but with more two-qubit gates on 16 and lower ESP, since at this scale layout and routing dominate. On ions it uses fewer two-qubit gates than level 3 on 11 circuits and more on none. On the 14 circuits BQSKit completed (up to 36 qubits) BQSKit level 3 has ×0.91 two-qubit gates on Heron against our ×0.98, but ×1.35 gates in total against ×0.82 (ions ×0.95/×0.63 against ×0.96/×0.53), with errors up to $3 \cdot 10^{-4}$ on at most 24 qubits and 22 wider outputs unchecked.

Table 6 Large circuits, ratios (12) relative to Qiskit level 3 (22 Heron and 25 ion circuits of up to 95 qubits). ESP ratios are over the 13 Heron and 25 ion circuits with level-3 ESP at least 10^{-3}, below which ESP is dominated by noise. Every output of our optimiser has an accepted replay certificate.

	IBM Heron			trapped ion		
pipeline	2q	all	ESP	2q	all	ESP
Qiskit level 1	×1.22	×1.13	×0.24	×1.40	×1.13	×0.851
TKET (FullPeepholeOptimise on L1)	×1.07	×1.96	×0.27	×1.39	×1.87	×0.828
ours, exact, from L1	×1.05	×0.78	×0.43	×0.96	×0.68	×1.033
ours, exact, from L3	×0.98	×0.81	×1.12	×1.00	×1.00	×1.011

Table 7 Logical level, Nam gate set, summed over 26 circuits, changes relative to the input. The last column counts circuits on which a tool is best or tied among the single tools. All outputs verified.

tool	gates	change	CNOT	change	best on
input	10,481	–	4,612	–	–
Qiskit level 3	10,391	-0.9%	4,511	-2.2%	0
TKET (FullPeepholeOptimise)	11,762	12.2%	4,489	-2.7%	0
QUESO (5 min per circuit)	9,765	-6.8%	4,465	-3.2%	4
VOQC	8,417	-19.7%	4,314	-6.5%	23
ours (exact up to phase, gate count)	9,480	-9.6%	4,426	-4.0%	2
VOQC, then ours	7,833	-25.3%	4,007	-13.1%	–

Certificates and run time.

171 of 176 outputs have an accepted certificate (Proposition 3), each with up to 544,922 steps, at most 6.8 s to check and residuals below 10^{-7}. A first checker rejected 7 certificates because it checked the phase repair as one block beyond its 12-wire limit. The current checker checks each patch separately and accepted all 2 re-run cases, and 5 were not re-checked. Median run times (Table C3) are seconds per circuit on the 60-circuit suite and minutes, up to 6.4 h, on the large suite in single-threaded Python, the main cost of the method at scale.

8.6 Logical-level comparison

Quartz [8], QUESO [9] and VOQC [13] are evaluated on the Nam gate set $\{H, X, \mathrm{RZ}, \mathrm{CNOT}\}$, all-to-all, with total gate count as objective, on 26 arithmetic and Toffoli circuits of [5]. Our optimiser runs unchanged with a database over this gate set and equality up to phase. Every tool starts from the decompositions distributed with QUESO. VOQC runs its full Nam pass sequence and QUESO its recommended rule sets with five minutes per circuit (one hour on 3 small circuits improves its total from 305 to 282). All 159 outputs are equal to their inputs up to phase.

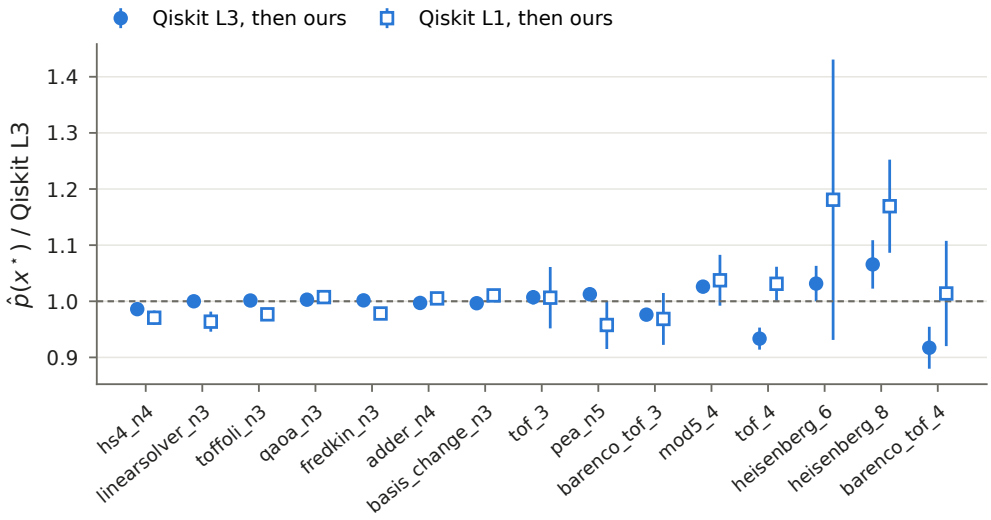


Fig. 5 ibm_marrakesh, paired ratio of Definition 7 relative to Qiskit level 3, mean and one standard deviation over 7 jobs. Circuits are sorted by level-3 gate count.

VOQC removes 19.7% of the gates and is best on most circuits, and ours removes 9.6%, ahead of QUESO, Qiskit and TKET (Table 7). VOQC merges rotations over phase polynomials arbitrarily far apart, while our windows see three qubits, and our windows find rewrites VOQC does not (`mod5_4`, 35 against 56 gates). Run on VOQC's output, our optimiser improves 21 of 26 circuits and gives the best result overall, 25.3% fewer gates and 13.1% fewer CNOTs than the input. Quartz was not run.

8.7 Real hardware

Definition 7 (Hardware metric) For a circuit c with deterministic ideal output $x^\star$, a pipeline P and job j, let $\hat{p}_{P,j}(c)$ be the fraction of shots returning $x^\star$, and $\bar{p}_P(c)$ its mean over jobs. The summary over N circuits is

$$\rho_{\mathrm{hw}}(P) = \Big(\prod_{c} \frac{\bar{p}_P(c)}{\bar{p}_{\mathrm{L3}}(c)}\Big)^{1/N}, \tag{13}$$

and per circuit we report the mean and standard deviation over j of the paired ratio $\hat{p}_{P,j}(c)/\hat{p}_{\mathrm{L3},j}(c)$, whose two factors share one calibration.

We ran $N = 15$ circuits of 3–8 qubits on ibm_marrakesh (IBM Heron r2), compiled by level 3, by level 3 followed by our optimiser and by level 1 followed by our optimiser, in 7 jobs of 4000 shots with the circuit order shuffled per job. Our optimiser on level 3 gives $\rho_{\mathrm{hw}} = 0.996$ (better on 8/15) and on level 1 1.015 (9/15), against 1.005 predicted by the noise model (Fig. 5). On these circuits it removes 12% of the gates but two-qubit gates on only 2, and the effect is below the spread between jobs.

Table 8 Layout-aware rewriting on IBM Heron. ESP (2) includes readout, ratios are geometric means relative to Qiskit's noise-aware level 3, and TVD is the mean exact noisy TVD including readout (5 circuits).

pipeline	ESP ratio	two-qubit gates (sum)	TVD
Qiskit L3 (coupling map only)	×0.63	493	0.569
Qiskit L3, noise-aware	×1.00	493	0.332
noise-aware + `qcr`	×1.01	485	0.325
noise-aware + `qcr` + relocation	×1.14	485	0.225
joint layout selection (`qcr`)	×1.14	471	0.222

8.8 Choosing the objective

We ran the optimiser with each cost of (1) and computed the exact TVD between ideal and noisy output distributions from density matrices (Heron calibrated errors of the actual qubits, ion depolarising noise at the model rates) on the 49 circuits with at most eight active qubits (Table C4). The calibrated cost is at least as good as the alternatives on both devices and beats Qiskit level 3 on 39/49 Heron circuits. Where no calibration exists, c_{w2} is a good proxy.

8.9 Layout-aware rewriting

On Heron we compare, for 8 circuits, Qiskit level 3 with and without the calibrated target (noise-aware layout), our optimiser on each, relocation (6) after the noise-aware pipeline, and joint layout selection over four seeds and two modes, with readout errors in ESP and in the noisy simulation.

Placement dominates (Table 8, Fig. C7). Qiskit's noise-aware layout raises ESP by 1/0.63 over the coupling-map-only layout, relocation after rewriting adds ×1.14 and joint selection ×1.14, with lower noisy TVD than the noise-aware baseline on 5/5 circuits. Joint selection chose a noise-aware candidate in only 50% of the circuits, so the best final layout is often not the one ranked best before optimisation.

9 Discussion and limitations

What the ground truth shows.

For discrete alphabets, local rewriting is rewriting on a Cayley graph, and its quality is governed by the database radius and the window through Theorem 5 and the cascade (7). Large hashed databases and convex windows close almost all of the gap to the optimum without using the group structure.

Exact rewriting and global passes.

Resynthesis (BQSKit) finds cheaper entangling structure inside blocks and rotation merging (VOQC) combines rotations far apart, both out of reach of stored rewrites on three-qubit windows. Exact local rewriting is faster, keeps the unitary including phase and removes what these passes leave (16% after BQSKit on Heron, the best of

all tools after VOQC). A practical pipeline runs a global pass first and exact local rewriting last.

Limitations.

(i) *Hardware evidence.* ESP and TVD come from calibration snapshots without idle errors or crosstalk, and the device run (Section 8.7) covers 15 small circuits, where the effect is below the run-to-run spread. (ii) *Global phase.* Exactness including phase costs 240 single-qubit gates over the Heron suite. With phase as metadata the all-gate ratio from level 1 would improve from $\times 0.838$ to $\times 0.825$. (iii) *Baselines.* Quartz was not run, QUESO had five minutes per circuit instead of one hour, other baselines use default settings, and on the large suite BQSKit covers only the 18 smallest circuits. (iv) *Scale.* Windows have at most three qubits, the code is single-threaded Python, and feed-forward is out of scope. Two 64-qubit Heron runs did not finish and 5 certificates were not re-checked. (v) *Ground truth* beyond Clifford is certified for two qubits and lengths up to 9. (vi) *Verification* is numerical up to 24 qubits and by replay certificates beyond, and exact only for Clifford and Clifford+T circuits.

Future work.

A theory of $w^\star$ would relate the cascade to Dehn presentations and local geodesics in groups [38, 39] and to complete rewriting systems [40]. Reaching the entangling structure that resynthesis finds without giving up exactness needs four-qubit databases with symmetry reduction, or resynthesised blocks with angles snapped to exactly realisable values and then verified. On hardware, the next step is repeated runs on circuits where rewriting removes two-qubit gates.

10 Conclusion

Making "equivalent" precise as a group action turns local rewriting into an exact, hardware-aware optimiser with provable guarantees. Built without using the structure of any gate set, it reaches proven optima on discrete three-qubit benchmarks and on planted Clifford+T circuits, improves Qiskit's level-3 output on real algorithms without adding two-qubit gates, scales to 95 qubits with independently checked certificates, and adds exact savings after BQSKit and VOQC. Exact ground truth, Clifford and beyond, is what makes such claims checkable.

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Declarations

- **Funding:** No funding was received for this work.
- **Competing interests:** The author declares no competing interests.
- **Ethics approval and consent to participate:** Not applicable.
- **Consent for publication:** Not applicable.

- **Data availability:** All raw results, logs and figure data are included in the repository below (`results/`).
- **Materials availability:** Not applicable.
- **Code availability:** The implementation (`qcr`, Apache License 2.0), every experiment script and the scripts that generate all tables, figures and numbers in this paper are available at https://github.com/arnavk23/qcr.
- **Author contribution:** A.K. conceived the study, developed the software, performed the experiments and wrote the manuscript.

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Appendix A Proofs

Proof of Proposition 1 By convexity, S can be moved to the front (a) or the back (b) of the circuit without changing $\mathrm{ev}(u)$. In case (a) the input factorises as $|0_W\rangle \otimes |0_{\bar{W}}\rangle$, so the final state changes only by the global phase $e^{i\varphi}$. In case (b) the final state changes from $|\psi\rangle$ to $(D \otimes I)|\psi\rangle$, and $|\langle x|(D \otimes I)|\psi\rangle|^2 = |\langle x|\psi\rangle|^2$ for every basis state x. Since the invariant is the output distribution of the current circuit, soundness composes. □

Proof of Proposition 2 Telescope over $u = u_0, u_1, \ldots, u_m = u'$. Each step changes one factor of a product of unitaries, $\mathrm{ev}(u_t) = A\,\mathrm{ev}(v_t)\,B$ and $\mathrm{ev}(u_{t-1}) = A\,\mathrm{ev}(S_t)\,B$ with unitary A, B, and $\|A(X - Y)B\| = \|X - Y\|$. □

Proof of Proposition 3 A swap changes ev by $A(\gamma'\gamma - \gamma\gamma')B$ with unitary A, B, of norm at most δ_t, so we may take $\lambda_t = 1$ for swaps. A replacement gives $\mathrm{ev}(u_t) = A\,\mathrm{ev}(v)\,B = \lambda_t \mathrm{ev}(u_{t-1}) + E_t$ with $\|E_t\| \le \delta_t$. Inductively $\mathrm{ev}(u') = \Lambda\,\mathrm{ev}(u) + E$ with $\Lambda = \prod_t \lambda_t$ and $\|E\| \le \sum_t \delta_t \prod_{s>t} |\lambda_s|$. Since $\mathrm{ev}(v)$ and $\mathrm{ev}(s)$ are unitary, $\big||\lambda_s| - 1\big| \le \delta_s$, hence $\|\mathrm{ev}(u') - \mathrm{ev}(u)\| \le |\Lambda - 1| + \|E\| \le \delta_0 + \sum_t \delta_t + O(\delta^2)$. □

Proof of Lemma 4 All generators map Paulis to Paulis under conjugation, so $\langle \Gamma \rangle \subseteq \mathcal{C}_n/U(1)$. The single-qubit generators generate $\mathcal{C}_1$, and an entangling Clifford together with $\mathcal{C}_1^{\otimes n}$ generates $\mathcal{C}_n$ [17]. Enumeration confirms the orders 24, 11,520 and 92,897,280 for $n = 1, 2, 3$. □

Proof of Theorem 5 Let u be (w, r)-reduced. If $|u| > D$, its first $D + 1$ letters form a factor f with $|\mathrm{ev}(f)| \le D \le r$ and $|f| = D + 1 \le w$, so the rule $f \to v$ with v a geodesic of $\mathrm{ev}(f)$ applies, a contradiction. Hence $|u| \le D < w$, so u is itself a factor of length at most w. If it were not geodesic, the rule $u \to v$ with v a geodesic of $\mathrm{ev}(u)$ and $|v| \le D \le r$ would apply. □

Proof of Proposition 6 If the whole word is a factor of length at most w and the database has unbounded radius, the word itself is replaced by a geodesic of its value. A fixed point of length $\ell(w) \le w$ is therefore a geodesic, and $w^\star$ is the smallest window for which this happens. □

Proof of Theorem 7 A Cayley graph is bipartite if and only if there is a homomorphism $\chi : G \to \{\pm 1\}$ with $\chi(\Gamma) = -1$. (a) For $n = 1$, $\mathcal{C}_1/U(1) \cong S_4$ and every $\pm\frac{\pi}{2}$ rotation is a 4-cycle, so $\chi = \mathrm{sgn}$ works. For $n = 2$ we verify bipartiteness exhaustively (no edge joins two elements at equal distance from the identity). (b) The Pauli group $\mathcal{P}_n$ is normal with quotient $\mathrm{Sp}(2n, 2)$. If χ existed, then $\chi(X_i) = \chi(\gamma)^2 = 1$ and likewise $\chi(Z_i) = 1$, so χ would factor through $\mathrm{Sp}(2n, 2)$, which is simple and non-abelian for $n \ge 3$. Hence $\chi \equiv 1$, contradicting $\chi(\Gamma) = -1$. □

Proof of Theorem 8 This is the standard L^2 bound for random walks on groups [41]. By Cauchy–Schwarz, $\|\mu_L - \pi_L\|_{TV} \le \frac{1}{2}\sqrt{|G|}\,\|\mu_L - \pi_L\|_2$, and the walk operator contracts the component orthogonal to the constants (and, in the bipartite case, to the sign character) by at least β per step. The limit of the expected length follows because $|\mathrm{ev}(u)| \le D$. □

Proof of Proposition 9 A geodesic of length $a+b$ splits into geodesics of lengths a and b, so $S(a+b) \subseteq S(a)S(b)$. Bounding $|B(d)|$ from above by these products bounds $\Pr[|g| \le d]$ from above, and $\mathbb{E}|g| = \sum_{d \ge 0} \Pr[|g| > d]$. □

Proof of Lemma 10 $\mathbb{D}[\omega]$ is closed under products and conjugation ($\overline{\omega} = -\omega^3$). If $V = e^{i\varphi}U$, the zero patterns agree and $V_{ij}\overline{V_p} = U_{ij}\overline{U_p}$. Conversely, $P(V) = P(U)$ at $(i,j) = p$ gives $|V_p| = |U_p|$, so $V_p = e^{i\varphi}U_p$ for some φ, and then $V_{ij} = U_{ij}\overline{U_p}/\overline{V_p} = e^{i\varphi}U_{ij}$. Uniqueness holds because $1, \omega, \omega^2, \omega^3$ is a $\mathbb{Z}$-basis of $\mathbb{Z}[\omega]$ and the smallest common exponent is unique. Divisibility by $\sqrt{2} = \omega - \omega^3$ is decided exactly by checking that all coefficients of $M\sqrt{2}$ are even. □

Appendix B Clifford-specific optimisers

For the alphabets of Section 6 the exact tables give an *exact-window* optimiser EW-w, which replaces every window of length at most w on at most three wires by an optimal word, and its convex version CW-w. CW-sem grows windows adaptively while gates commute into them.

Table B1 Mean gate counts on the 3-qubit benchmarks (300 gates, 20 circuits).

	optimum	original	EW-8	EW-16	Qiskit L3
ion	10.4	47.1	28.3	10.4	127.2
IBM-style	12.2	36.8	51.1	12.2	104.0
time per circuit (ion)	–	51 s	0.54 s	0.24 s	<0.1 s

Findings.

(1) On these benchmarks the original method finishes 4.5× (ion trap) and 3.0× (IBM-style) above the exact optimum. (2) EW-w reaches the *proven* optimum on all 20 circuits once $w \ge 12$ (ion trap) or $w \ge 16$ (IBM-style), in a fraction of a second. This is the cascade of Prop. 6. The transition is sharp, between $w = 8$ and $w = 12$ (ion trap) or $w = 16$ (IBM-style). (3) The grid (Fig. 3) shows that the *database radius*, not the window or the search strategy, is the binding constraint. At $r = 5$, enlarging the window from 8 to 16 lowers the mean excess only from 46 to 43 gates. At $w = 8$, raising r from 5 to ∞ lowers it from 46 to 18, and for $w \ge 12$ with $r \ge 8$ the excess is at most 2 gates (ion trap, and from 72 to ≤ 8 for the IBM-style set). Theorem 5 predicts both requirements, windows long enough *and* a database deep enough. The original method is limited by the second.

Findings.

(1) *Convexity matters more than anything else at $n \ge 4$.* With the same exact 3-qubit oracle, contiguous windows (EW-16) give 110.3 (ion trap) and 130.9 (IBM-style) gates, no better than the published method. With DAG-convex windows (CW-24) the same

Table B2 4-qubit benchmark (300 gates, 100 circuits). Original (orig.) and Qiskit level 3 use the first 30 circuits. Published (publ.) numbers are from the journal version of [1]. BQSKit level 4 is our run on the first 30 circuits, with the same gate set at continuous angles (approximate, all-to-all).

	bound	CW-sem	CW-24	EW-16	orig.	publ.	Qiskit	BQSKit
ion, total	≥11.1	45.6	56.7	110.3	78.8	111	164.9	120.6
ion, RXX	–	–	27.6	44.2	33.3	43	47.3	27.6
IBM-style, total	≥12.9	49.3	59.1	130.9	76.0	107	140.9	165.2
IBM-style, CZ	–	–	27.1	47.1	32.4	43	47.3	35.1

oracle gives 56.7 and 59.1 gates, i.e. 49% and 45% fewer than the published results, in about 5.6 s per circuit. (2) Two-qubit counts drop from 43 (published) to 27.6 RXX and 27.1 CZ, although CW optimises total length, not two-qubit count. On the first 30 circuits, where we also ran BQSKit level 4 (Table B2), CW-24 averages 27.0 RXX against BQSKit's 27.6 (fewer on 16, more on 13) and 28.1 CZ against 35.1 (fewer on 25, more on 4), with ×0.46 and ×0.37 of BQSKit's total gates and exact outputs. The journal reports 28 and 37 two-qubit gates for BQSKit. (3) The certified lower bounds (11.1, 12.9) leave a gap of at most a factor of about 4 (45.6/11.1 and 49.3/12.9). Closing it, either by tightening the bound (deeper spheres, meet-in-the-middle) or by better upper bounds (4-qubit oracles via symmetry reduction), is an open problem that our framework makes precise.

Appendix C Additional figures and tables

Table C3 Median run time per circuit in seconds. Qiskit level 1 was not timed on the 60-circuit suite. On the large suite BQSKit covers only the 18 smallest circuits, and part of its run shared the machine with an unrelated job. The large suite ran several circuits at a time on eight cores.

	60-circuit suite		large suite	
pipeline	Heron	ion	Heron	ion
Qiskit level 1	–	–	0.11	0.04
Qiskit level 3	0.03	0.01	0.12	0.05
TKET	0.30	0.86	8.2	9.6
BQSKit level 2 (8 workers)	14.7	4.8	132	34.2
BQSKit level 3 (8 workers)	53.6	11.2	842	138
ours, exact, from L1	10.4	3.0	853	187
ours, exact, from L3	6.0	4.8	117	57.1

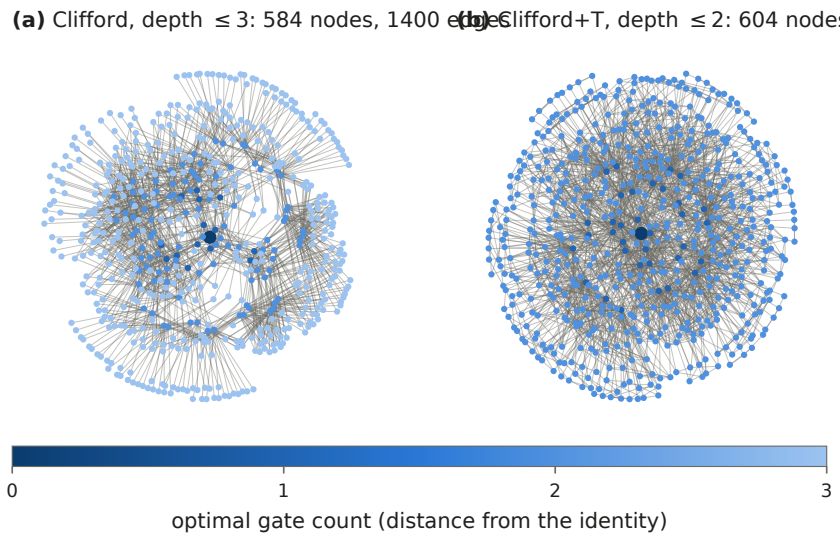


Fig. C1 Compute graphs of comparable size, nodes coloured by exact distance from the identity. (a) Two-qubit Clifford group, ion alphabet. Many short words coincide and the ball closes up. (b) Two-qubit Clifford+T. Almost every word is new.

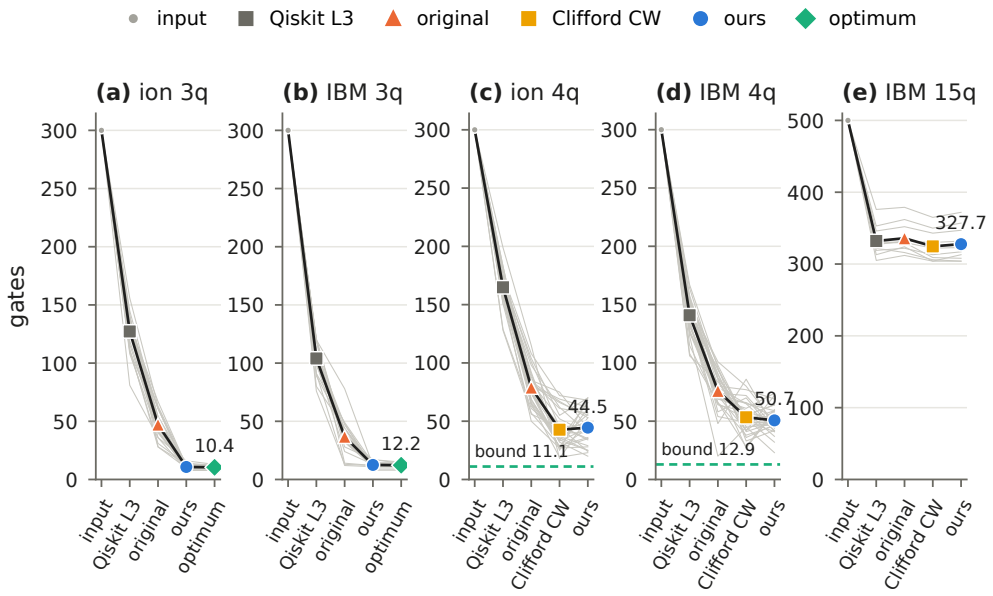
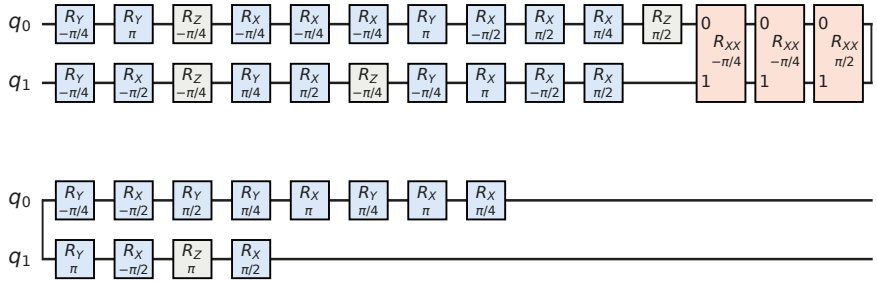


Fig. C2 From the input to the output of every method on the discrete benchmarks. Grey lines are individual circuits (paired by seed), the black line is the mean and the number is the final mean. The dashed line at 4 qubits is the certified lower bound of Proposition 9.

(a) planted Clifford+T circuit, 36 gates



(b) general optimiser, 5 gates (certified optimum 5)

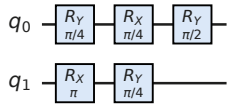


Fig. C3 A planted Clifford+T circuit and the output of the general optimiser, exactly equal over $\mathbb{D}[\omega]$.

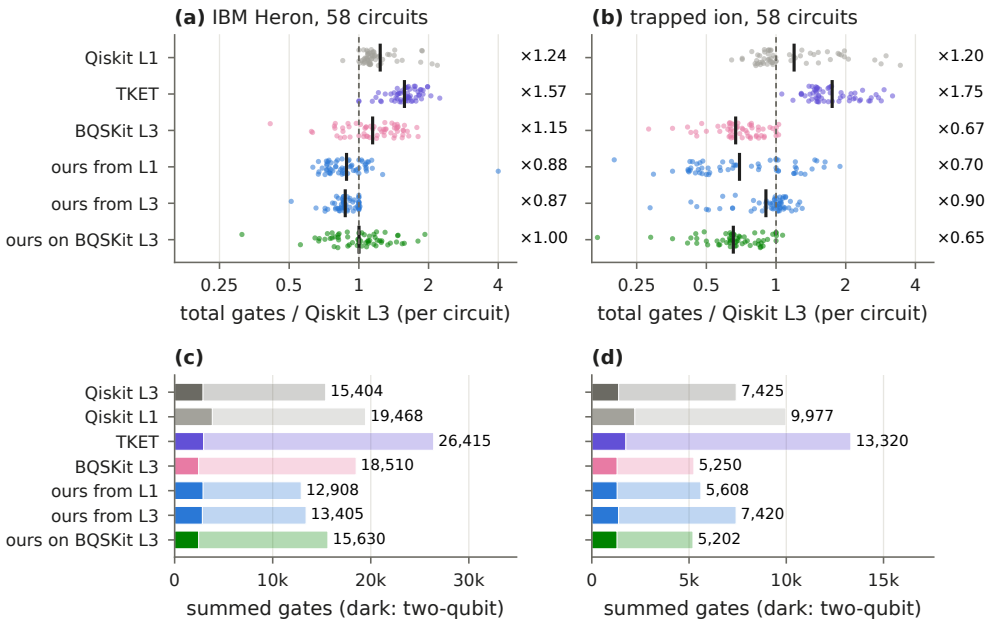


Fig. C4 Hardware suite. (a, b) Total gates per circuit relative to Qiskit level 3 (dots) and geometric means (bars and labels). (c, d) Summed gates over all circuits, two-qubit gates dark.

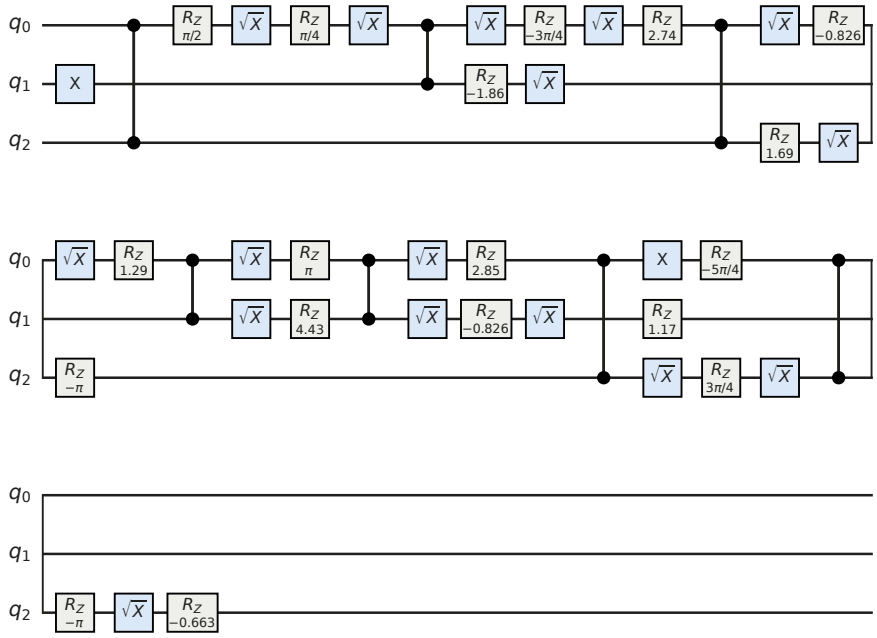
Table C4 Mean noisy-simulation TVD by objective (lower is better).

	Qiskit L1	Qiskit L3	ours, c_{count}	ours, c_{w2}	ours, c_{cal}
Heron	0.115	0.137	0.096	0.088	0.088
ion	0.0157	0.0091	0.0092	0.0084	0.0084

Table C5 Hashed databases of the general optimiser (classes per equivalence notion).

device	topology	notion	depth	classes	build time
Heron	line	up to phase	9	6.31M	6 min
Heron	line	output-aware	10	0.34M	<1 min
Heron	line	input-aware	14	1.41M	<1 min
ion	triangle	up to phase	6	28.4M	38 min
ion	triangle	output-aware	6	7.36M	27 min
ion	triangle	input-aware	7	31.6M	23 min
ion ($\pm\frac{\pi}{2}$)	triangle	up to phase	9	2.93M	7 min
IBM ($\pm\frac{\pi}{2}$)	triangle	up to phase	10	1.84M	2 min

(a) after Qiskit level 3, 43 gates, 7 CZ



(b) after `qcr`, 36 gates, 7 CZ

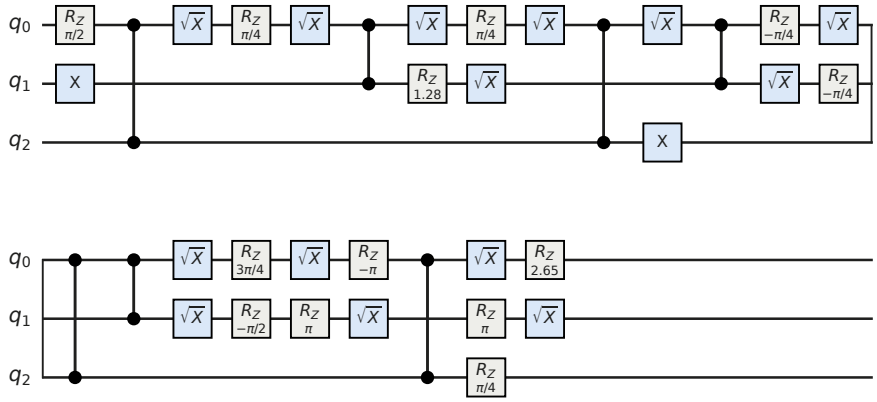
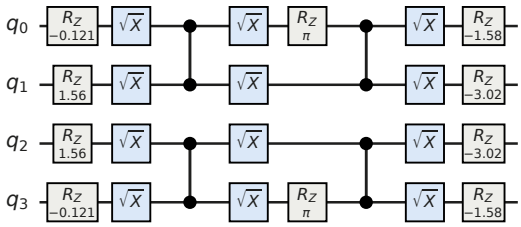


Fig. C5 QASMBench `toffoli_n3` on IBM Heron. The outputs are equal including global phase. Blue boxes are pulsed gates and grey boxes virtual RZ.

(a) after Qiskit level 3, 26 gates, 4 CZ



(b) after `qcr`, 20 gates, 4 CZ

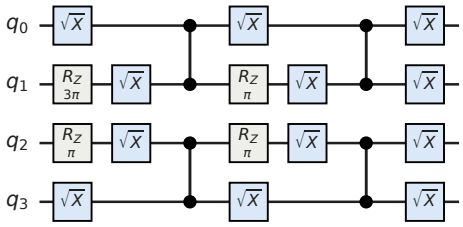


Fig. C6 QASMBench `hs4_n4` on IBM Heron, drawn as in Fig. C5.

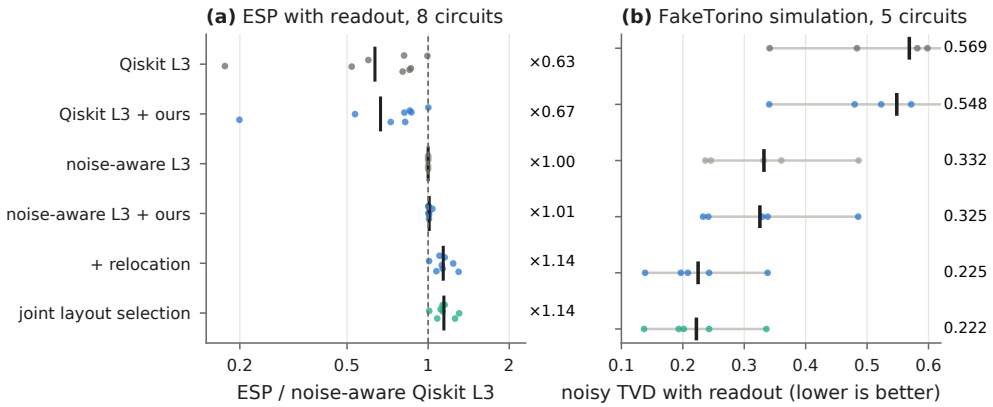


Fig. C7 Layout-aware rewriting on IBM Heron (FakeTorino). (a) ESP including readout relative to Qiskit's noise-aware level 3 (dots are circuits, bars and labels geometric means). (b) Exact noisy TVD including readout for circuits with at most eight active qubits (bars and labels are means).